

AYANANTA CHOWDHURY



AI Enterprise Knowledge Systems Architect | Data Governance Manager / Ontology Engineer

Architecting Knowledge for AI-Native User Assistance in Enterprise Software

Mobile: 9730821120

Email: ayananta.chowdhury@gmail.com

LinkedIn: <https://www.linkedin.com/in/ayanantachowdhury/>

Website: <https://ayananta.github.io/Digital-Resume/> **Location:** India

Professional Summary

AI-Native User Assistance Architect with 15+ years in Technical Documentation (product/support and maintenance/API documentation/UX writing), including 4+ years in developing AI-ready enterprise information architecture. During my last role at HCL Technologies, I transformed the knowledge base of a SaaS ERP solution, SAP Business ByDesign, into a computational format for machines to read, reason over, and act upon, enabling real-time assistance to users within product workflows. Also, built faceted-search function into the company knowledge portal of a tech start-up, specializing in Hyperlocal CDN solutions.

Work Experience

HCL Technologies | Senior Technical Manager – User Assistance (Mar 2023 – Apr 2025, Remote) Fulltime

Impact: As an Information Architect, modeled the product documentation of a SaaS ERP solution from SAP to enable AI-native user assistance. In addition to making the documentation machine-readable, enhanced overall content reuse by 11%.

Key Contributions

- Framed competency questions to define scope, established business goals, and identified information sources to build an OWL ontology for a SaaS ERP solution from SAP.
- Used NLP tools to extract noun phrases from information sources. Used a clustering tool to group related terms, then analyzed the clusters to identify hypernyms (umbrella terms) and hyponyms (specific terms) among the key concepts.
- Organized hypernyms into a logical hierarchy as top-level categories and grouped hyponyms beneath their appropriate parents, forming the concept taxonomy that would serve as the ontology's class structure.
- Specified “preferred,” “admitted,” and “deprecated” terms in Acrolinx for real-time authoring guidance and error-flagging, supporting controlled vocabulary governance, data quality consistency, and backward compatibility.
- Linked the top-level concepts to their immediate next-level ones via semantic relationships to sketch a UML class diagram, which acted as a prototype for SMEs to validate the accuracy of the knowledge model.
- Designed a rigorous OWL ontology by linking resources through more specific relationships, stipulating applicable quantifiers, hasValue, cardinality, and constraint restrictions for every class, improving semantic integrity.
- Chunked the structured documentation to represent every user intent or concept as discrete knowledge units, assigning each unit a unique ID and externalizing content with rich semantic metadata for improved traceability.

Quincus | API Technical Writer (Oct 2022 – Nov 2022, Remote) Contractual

Documented API endpoints, resources, methods, authentication protocols, parameters, & headers for a logistics platform.

SugarBox Networks | Technical Communications Manager (Aug 2020 – Sept 2022, Remote) Fulltime

Impact: Built faceted-search function into a knowledge base for the company's Hyperlocal CDN solution. Additionally, by providing tailored user guidance, the Tech Communication team reduced stakeholder onboarding time by 31.4%.

Key Contributions

- Created a terminology of concepts associated with Content Delivery Network (CDN) to build a controlled vocabulary.
- Ordered the broadest concepts logically to build a taxonomy.
- Developed a semantic network of CDN-related concepts. Created a concept scheme, the scheme's top concept, and the top concept's related concepts and linked them—asserting their semantic relationships through SKOS object properties.
- Recorded annotations for skos:prefLabel, skos:altLabel, skos:hiddenLabel to support standardized metadata governance.
- Created a subject scheme map to bind controlled vocabulary values to DITA attributes, enabling automated content filtering, uniform metadata usage, and validation of attribute values. Ran Schematron rules to enforce consistent metadata usage.
- Created embedded assistance and walkthrough videos for the company's CDN solution. Reviewed the contextual help (a Configuration Guide) authored by team members for compliance with guidelines. Additionally, conducted performance and stakeholder escalation management, hired resources, and analyzed content effectiveness and user engagement metrics.

HCL Software (Product Div) | Technical Writing Senior Consultant (Nov 2018 – Mar 2020, Pune) Fulltime

Impact: Streamlined the documentation workflow to align with the CI/CD release cycle for a data engineering platform.

Key Contributions

- Led the doc team, mentoring members on topic-based structured authoring; managed doc task workflow in JIRA.
- Audited existing content by running TEXTML queries to identify redundant content, and then implemented content reuse.
- Drafted hotfix docs & release notes and reviewed content on new feature enhancements for a data-engineering solution.
- Added parameter descriptions in the Open API Specification file and wrote the conceptual part of the API documentation.

Sungard Discovery Services | Specialist, Technical Documentation (May 2017 - Jul 2017, Pune) Fulltime

- Created video tutorials explaining the capabilities of the company's disaster recovery product.

Persistent Systems | Information Development Team Lead (Nov 2013 – May 2017, Pune) Fulltime

- Produced demo videos of medical device MVPs for potential investors (life sciences & healthcare domain).
- Coordinated with cross-functional teams of Merck & Co. for ITIL-based change management, using ServiceNow.
- Collaborated with the VMware on-site team to author contextual help, including a quick-start-guide, and wrote UI microcopy.
- Performed usability testing to verify accuracy of content that was authored by team members.

Rolta India | Senior Technical Writer - Product Documentation (Aug 2012 – Jul 2013, Mumbai) Fulltime

- Explained the new features of a GIS software, following content accessibility standards in an Agile environment.

Polaris Software Lab | Technical Writer (Oct 2010 – Aug 2012, Mumbai) Fulltime

- Co-authored user and installation guides following WCAG accessibility standards for a banking solution that processed cross-border transactions via SWIFT messaging, ensuring accurate fund settlements.

Connectiva Systems | Technical Writer (Feb 2008 – Sept 2010, Kolkata) Fulltime

- Built storyboards for eLearning videos (telecom domain), edited corporate training PPTs, and updated existing user guide.

Compare Infobase | Content Writer (Jan 2007– Dec 2007, Kolkata) – Wrote SEO content for travel websites.

Education: Post-graduate in Mass Communication and Journalism, Calcutta University | 2005–2006

Competencies | Skills

KNOWLEDGE MODELING

- Rich Taxonomies - SKOS
- Ontologies – UML, OWL
- Simple Taxonomies – RDF, RDFS
- Knowledge Graph Model - RDF
- Validation language – SPARQL, SHACL
- Serialization Format – Turtle

DATA GOVERNANCE & AI TRUST

- Controlled Vocabulary Governance
- AI Readiness & AI Trust
- Provenance & Traceability
- Metadata Governance

ARTIFICIAL INTELLIGENCE

- RAG and Langchain
- Machine Learning
- Generative AI
- Agentic AI
- Natural Language Processing

CONTENT STRATEGY AND DEVELOPMENT

Study and analyze the target audience to create user personas, journey maps, and workflow diagrams. Define KPIs for successful user assistance, measure help success, and create a master documentation list with editorial timelines. Develop multimodal help content to assist users in setting up, operating, and maintaining enterprise applications using video tutorials, guides, and visual learning resources such as diagrams, infographics, and workflows. Author release notes, hotfix doc, FAQs & troubleshooting, and training materials. Produce SEO-optimized (E-E-A-T) marketing content and edit SOPs, runbooks, process documentation, and functional & technical specifications to align with Microsoft Manual of Style (MSTP) guidelines. Embed assistance (microcopy, UX writing) and walkthrough videos (digital guided tours) on UI.

API DOCUMENTATION

Document each endpoint's parameters, headers, and JSON/XML request/response payloads.

Tools

IxiaSoft CCMS, SpaCy, Open Refine, Carrot2, Protégé, Blazegraph | Git, Markdown | Camtasia, Visio, Napkin, HeyGen, and Lucidchart | Google Analytics, Hotjar | DITA XML-based Authoring Systems, RoboHelp / MadCap Flare, Confluence, Docs, MS Office | WalkMe | JIRA, ServiceNow | Swagger, Postman | Cursor AI for vibe coding

Certifications

SKOS Taxonomy | Ontology Development | RDF & SPARQL | API Documentation | Git | Advanced AI Concepts | UX Writing